Name: _____

TOPIC TEST 1

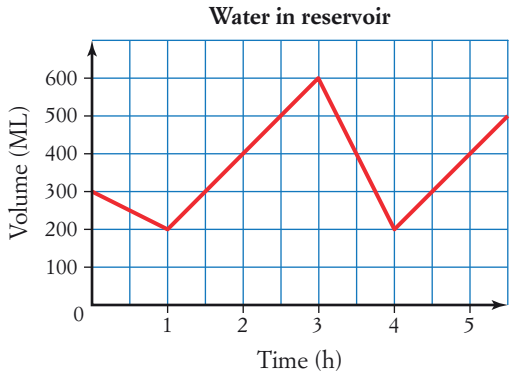
Chapter 7: Rates of change

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 5 short answer questions (22 marks)
- Part C: 2 analysis questions (14 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 This graph shows the amount of water in a reservoir over time.



The average rate of change of the amount of water from 0 to 2 hours was:

A 150 ML/h

B 200 ML/h

C 50 ML/h

D 130 ML/h

E 100 ML/h

[1 mark]

© Cengage Learning Australia Pty Ltd 2013

2 From the graph in question **1**, the instantaneous rate of change of the water level at 2 hours was:

A 150 ML/h

B 200 ML/h

C 50 ML/h

D 130 ML/h

E 100 ML/h

[1 mark]

3 A stone drops into deep water below a cliff that is 10.2 m above the water. The stone is travelling at a speed of 14.8 m/s when it hits the water. After travelling 2.1 m down through the water in 0.15 seconds it has slowed to 13.2 m/s. If the rate of change of the speed remains the same, how long after hitting the water will it be at 6.8 m/s?

A 0.75 s

B 0.6 s

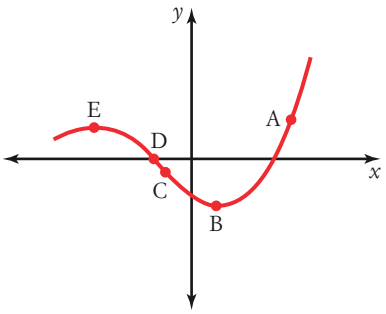
C 1.6 s

D 1.07 s

E 0.9 s

[1 mark]

4 The diagram below shows the graph of a function.



At which point on the graph is the rate of change the greatest?

[1 mark]

5 The rate of change of $f(x) = x^3 - 3x + 3$ at $x = 2$ is approximately:

A 5

B 8

C 12

D 6

E 9

[1 mark]

Part B

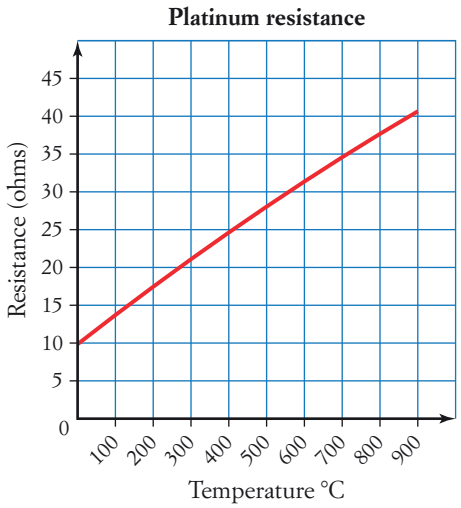
5 short answer questions
22 marks
Show your working where appropriate.

1 In 2002 the price of a 250 L refrigerator was \$429, but by 2012 the price of a comparable 250 L refrigerator had risen to \$609. What was the average rate of price increase?

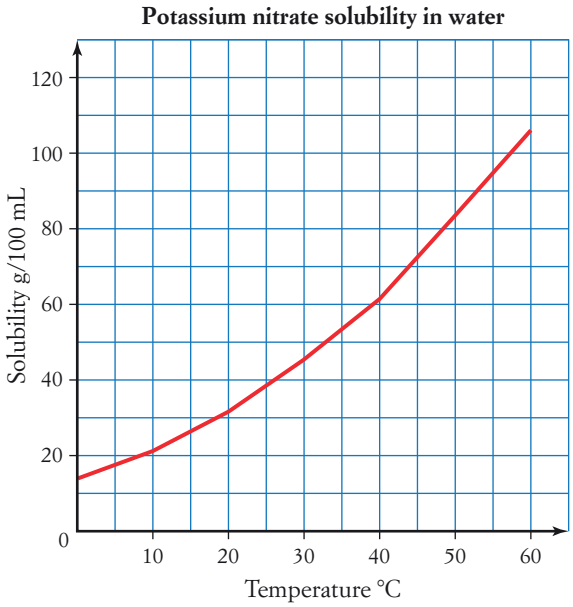
[2 marks]

2 The graph below shows the resistance of a platinum wire at different temperatures. Find the average rate of change of resistance between 200° and 600°.

[4 marks]



3 The graph below shows the solubility of potassium nitrate at different temperatures. Use a tangent to find the rate of change of the solubility at 30 °C.



[7 marks]

4 Find the gradient of $f(x) = \dfrac{x^2 - 1}{x^2 + 2}$ at $x = 3$ using a tangent (on your CAS calculator).

[4 marks]

5 Use $\delta x = 0.06$ to find the approximate gradient of $y = x^2 - 5$ at $x = 2$ using a secant around the point.

[5 marks]

© Cengage Learning Australia Pty Ltd 2013

Part C

2 analysis questions
14 marks
Show your working where appropriate.

1 Find approximate rates of change for the function $f(x) = 5 - 2x^2 + x$ at values of x from 0 to 4 to suggest a function for the rate of change of $f(x)$.

[7 marks]

2 Find approximate rates of change for the function $f(x) = \sqrt{x + 2}$ at values of $x = -1, 2, 7$ and 14 to suggest a function for the rate of change of $f(x)$.

[7 marks]
Total marks: 41

© Cengage Learning Australia Pty Ltd 2013

Answers

Part A **[1 mark each]**

1 Average rate of change $= \dfrac{400 - 300}{2 - 0} = 50$ ML/h

$\therefore$ **C** **[1 mark]**

2 The line at 2 hours passes through (1, 200) and (2, 400), so the rate of change $= \dfrac{400 - 200}{2 - 1} = 200$ ML/h

$\therefore$ **B** **[1 mark]**

3 Rate of change of speed $= \dfrac{13.2 - 14.8}{0.15} = -10\frac{2}{3}$ m/s/s

Desired change of speed $= 6.8 - 14.8 = -8$ m/s

Time for change $= \dfrac{-8}{-10\frac{2}{3}} = \frac{3}{4} = 0.75$ s

$\therefore$ **A** **[1 mark]**

4 The greatest rate of change is where the graph is steepest.

$\therefore$ **A** **[1 mark]**

5 Choosing $\delta x = 0.2$, rate of change $\approx \dfrac{(2.1^3 - 3 \times 2.1 + 3) - (1.9^3 - 3 \times 1.9 + 3)}{0.2} = 9.01$

$\therefore$ **E** **[1 mark]**

Part B

1 Average rate of increase of the price $= \dfrac{609 - 429}{2012 - 2002}$ **[1 mark]**

$= \$18$/year **[1 mark]**

[2 marks]

2 Resistance at 200 °C = 17.5 ohm **[1 mark]**

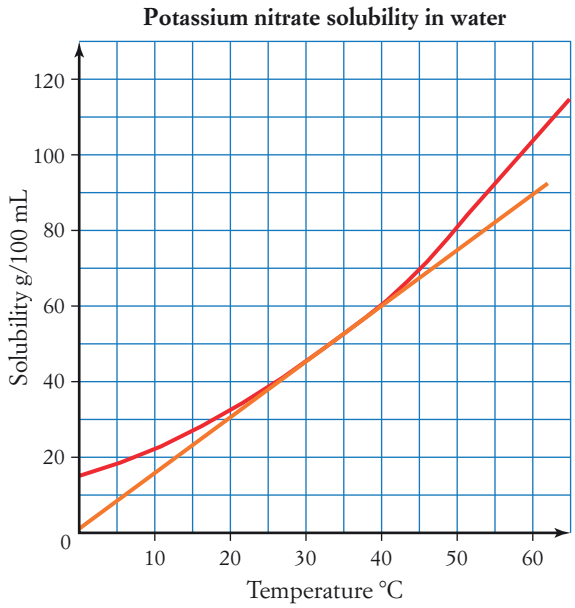
Resistance at 600 °C = 31.5 ohm **[1 mark]**

Rate of change of resistance $= \dfrac{31.5 - 17.5}{600 - 200}$ **[1 mark]**

$= 0.035$ ohm/°C **[1 mark]**

[4 marks]

3 Draw in a tangent at 30 °C.



[2 marks for reasonably accurate tangent]

The tangent goes through (0, 0) and (50, 74). **[1 mark each]**

Rate of change of solubility $= \dfrac{74 - 0}{50 - 0}$ **[1 mark]**

≈ 1.5 g/100 mL/°C **[1 mark for figure, 1 for units]**

[7 marks]

4 Enter the function as a graph. **[1 mark]**

Use trace and zoom-in to put a point at (3, 0.727). **[1 mark]**

Draw a tangent and measure the slope to get 0.149. **[2 marks]**

[4 marks]

5 Calculate the values at 1.97 and 2.03 to get (1.97, −1.1191) and (2.03, −0.8791). **[1 mark each]**

Then $\delta y = -0.8791 - (-1.1191) = 0.24$ **[1 mark]**

So the gradient is $\dfrac{\delta y}{\delta y} = \dfrac{0.24}{0.06}$ **[1 mark]**

$=$ (approximately) 4 . **[1 mark]**

[5 marks]

Part C

Students are expected to use CAS calculators efficiently.

1 At $x = 0$, the approximate rate of change is 1. **[1 mark]**

At $x = 1$, the approximate rate of change is −3. **[1 mark]**

At $x = 2$, the approximate rate of change is −7. **[1 mark]**

At $x = 3$, the approximate rate of change is −11. **[1 mark]**

At $x = 4$, the approximate rate of change is −15. **[1 mark]**

This suggests that the function for the rate of change is $y = 1 - 4x$ **[2 marks]**

[7 marks]

2 At $x = -1$, the approximate rate of change is $0.5 = \dfrac{1}{2}$ **[1 mark]**

At $x = 2$, the approximate rate of change is $0.25 = \dfrac{1}{4}$ **[1 mark]**

At $x = 7$, the approximate rate of change is $0.167 \approx \dfrac{1}{6}$ **[1 mark]**

At $x = 14$, the approximate rate of change is $0.125 = \dfrac{1}{8}$ **[1 mark]**

The calculations all involved $\sqrt{x + 2}$, and for −1, 2, 7 and 14 these are 1, 2, 3, and 4. **[1 mark]**

This suggests that the rate of change function is $y = \dfrac{1}{2\sqrt{x + 2}}$ **[2 marks]**

[7 marks]

© Cengage Learning Australia Pty Ltd 2013